

Nuclear & Clean-Energy Innovation Opportunity Intelligence Platform

Executive Brief — One Page | Karan Gill | July 2026

Purpose

A decision-support platform that identifies, scores, and prioritizes 14 emerging nuclear and clean-energy technologies for research, investment, partnership, or commercialization — built as an independent portfolio project demonstrating the analytical workflow used by innovation and business-development teams in the nuclear sector.

Method

- Structured a 14-technology dataset from public sources (IAEA, OECD NEA, CNSC, DOE, NWMO, peer-reviewed literature), with every field tagged VERIFIED / ESTIMATED / ASSUMPTION.
- Built a transparent, weighted scoring model (10 criteria, adjustable weights, min-max normalization) with a full sensitivity analysis — the #1 ranking is stable under $\pm 20\%$ perturbation of every individual weight.
- Applied an NLP pipeline (TF-IDF + KMeans clustering, cosine-similarity semantic search) to a real, source-linked research corpus, validated at 100% category-purity against known technology labels.
- Built an interactive dashboard (Power BI-ready data model + build guide, plus a working HTML preview) covering executive overview, ranked portfolio, TRL-vs-market matrix, cost-vs-impact, and risk analysis.

Top-Ranked Opportunities

Rank	Technology	Score
1	AI/ML for Predictive Maintenance	77.2
2	Advanced Sensor Technologies	65.3
3	Small Modular Reactors (SMRs)	63.8
4	Robotic Nuclear Inspection	63.6
5	Digital Twins for Nuclear Plants	63.5

Recommendation

Proceed with a scoped, low-capital proof-of-concept on AI/ML-based predictive maintenance (full business case in `business_case_ai_predictive_maintenance.docx`), conditional on confirming historical maintenance/sensor data availability — the one material unknown a desk-based analysis cannot resolve.

What This Demonstrates

- Investigating emerging technologies and synthesizing complex technical information into a defensible ranking.
- Market/competitive analysis and commercialization judgment grounded in named, checkable public sources.
- Applied AI/ML and data analytics used to solve a real triage problem, not added for its own sake.
- Clear, honest separation of verified fact, estimate, and assumption throughout — including where the project's own research method hit real constraints (see `technical_white_paper.docx`, Limitations).

Full repository, dashboard, white paper, market assessment, roadmap, and risk register available on request.